Lesson 18: Complex Correlation

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 6.4: Complex Correlation.

In this lesson

You will learn how researchers measure several variables at once and analyse them with a correlation matrix, factor analysis, partial correlation and regression, and how far these methods can go towards causal conclusions. You will practise conditional sentences that interpret possible results ("this would suggest that …").

Learning goals

After this lesson you should be able to:

- Read and build a correlation matrix.
- Explain what factor analysis does and what it does not tell us.
- Explain statistical control with partial correlation, and interpret partial correlations.
- Distinguish predictor and outcome variables and explain what multiple regression adds.
- Interpret hypothetical results with hedged conditional sentences.

Key language

Term or phrase	Meaning	Example
correlation matrix	a table that shows the correlations among several variables	need for cognition, intelligence, social desirability, dogmatism
factor analysis	a technique that organises many variables into clusters (factors) with strong correlations within a cluster and weak correlations between clusters	the Big Five personality factors
factor loading	how strongly a variable is related to a factor (shown in the book's table)	jazz: .83 on *Reflective and Complex*
statistical control	accounting for third variables by measuring them and including them in the analysis	controlling for socioeconomic status (SES)
partial correlation	the relationship between two variables while statistically controlling for one or more third variables	violent TV and aggression, controlling for SES
regression	a technique that predicts one variable from another	predicting GPA from IQ

Term or phrase	Meaning	Example
predictor / outcome (criterion) variable	the variable used to make the prediction / the variable being predicted	IQ / GPA
regression weight (*b*)	how much the outcome variable changes for each one-unit change in a predictor	`Y = b1X1`
multiple regression	predicting one outcome from several predictors, showing the separate contribution of each	income and health predicting happiness
over and above	in addition to what others already explain	a contribution over and above the other predictors
this would suggest that	a hedged interpretation of a possible result	That would suggest that SES is indeed a third variable.
it is important to note that	introduces a key limitation	It is important to note that partial correlation does not solve the directionality problem.

Rules and explanations

The ideas

Measuring several variables

Complex correlational research measures several variables (binary or continuous) and assesses the relationships among them.

- **Radcliffe and Klein (2002):** more optimistic middle-aged adults were healthier (they exercised more and had lower blood pressure), knew about heart attack risk factors, and correctly believed their own risk to be lower than their peers'.
- **Jouriles et al. (2009):** because measures of physical aggression gave highly skewed distributions, the researchers turned them into binary measures (0 = did not occur, 1 = did occur). Adolescents who experienced physical aggression were moderately likely also to have experienced psychological aggression, and psychological aggression was related to psychological distress.

The correlation matrix

Cacioppo and Petty (1982) tested their Need for Cognition Scale (how much people like and value thinking) by correlating it with intelligence, socially desirable responding and dogmatism:

	Need for cognition	Intelligence	Social desirability	Dogmatism
Need for cognition	—			
Intelligence	+.39	—		

	Need for cognition	Intelligence	Social desirability	Dogmatism
Social desirability	+.08	+.02	—	
Dogmatism	−.27	−.23	+.03	—

Only half is filled in, because the other half would repeat the same values; the diagonal has dashes, because a variable always correlates +1.00 with itself. The pattern fitted the researchers' ideas, which supports the validity of the new measure.

Factor analysis

Factor analysis organises many conceptually similar variables into a smaller number of clusters; each cluster is interpreted as several measures of one underlying construct (a **factor**). Examples: mental tasks form a mathematical and a verbal intelligence factor; warmth, gregariousness, activity level and positive emotions form extraversion, one of the Big Five. Rentfrow and Gosling asked more than 1,700 students to rate 14 music genres and found four factors: *Reflective and Complex* (blues, jazz, classical, folk), *Intense and Rebellious* (rock, alternative, heavy metal), *Upbeat and Conventional* (country, soundtracks, religious, pop) and *Energetic and Rhythmic* (rap/hip-hop, soul/funk, electronica). (The book dates this study 2008 in the text and 2003 in the table; the article was published in 2003.)

Two points:

1. **Factors are not categories.** People are not *either* extraverted *or* conscientious; factors operate independently.
2. **Factor analysis only reveals structure.** Researchers must interpret and label the factors and explain their origin (for example, the Big Five appear to be controlled by different genes).

Exploring causal relationships: partial correlation

Correlational research cannot establish causation unambiguously, but it can **rule out** some interpretations by **statistical control**: measuring possible third variables and including them in the analysis. In **partial correlation**, researchers examine the relationship between two variables while controlling for a third.

The book's example: violent television viewing and aggression correlate +.35 (moderate). The researcher also measures SES and controls for it:

Partial correlation (controlling for SES)	Interpretation
+.34	The relationship is largely independent of SES; SES is not a third variable driving it.
+.03	SES is indeed a third variable driving the relationship.
+.20	SES accounts for some, but not all, of the relationship.

Limits: partial correlation does not solve the **directionality problem**, and there may be other third variables that the researcher did not measure.

Regression

Once a relationship is established, **regression** predicts one variable from another (IQ predicting GPA). The book gives the simple form `Y = b1X1`: the predicted score equals the person's score on the predictor multiplied by the **regression weight**. (Full regression equations usually also contain a constant, the intercept; the book's simplified form leaves it out.) **Multiple regression** uses several predictors:

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Y = b1X1 + b2X2 + b3X3 + ... + biXi
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Its advantage: it shows whether a predictor contributes **over and above** the others. Income and health are related to each other, so either might seem to predict happiness only because of the other; multiple regression shows the contribution of each with the other statistically controlled. Such research has shown that both make extremely small contributions to happiness, except in severe poverty or illness (Diener, 2000).

Purely correlational approaches can only show patterns "consistent with some causal interpretations and inconsistent with others".

Language focus

Interpreting possible results with conditionals. The section imagines several results and interprets each with *if* + past tense, *would* + verb:

- "If she found that the partial correlation … was +.34, that **would suggest that** the relationship … is largely independent of SES."
- "If, however, she found that statistically controlling for SES reduced the magnitude of the correlation from +.35 to +.20, then this **would suggest that** SES accounts for some, but not all, of the relationship."

It then states limits with a fixed frame: "**It is important to note that** while partial correlation provides an important tool …, researchers using this technique are still limited in their ability to arrive at causal conclusions."

Use *would suggest* (not *proves*) for planned or hypothetical results, and keep *if* + past simple, not *if* + *would*.

> **Note for Arabic speakers** - Do not put *would* in the *if* clause: write "If the correlation **dropped** to +.03, this **would** suggest …", not "If the correlation would drop …".

Practice

1. Reading questions

1. What did Radcliffe and Klein (2002) find about optimistic adults?

2. Why did Jouriles and colleagues turn their aggression measures into binary variables?
3. In the Cacioppo and Petty matrix, what is the correlation between need for cognition and dogmatism? Which correlation is the weakest?
4. What does factor analysis do? Give the book's extraversion example.
5. In the music table, which factor does jazz load on, and how highly? Which genre has loadings of .40 or larger on two factors?
6. What are the two additional points about factor analysis?
7. Why can partial correlation still not establish causation?
8. What do regression weights indicate, and what is the advantage of multiple regression?

2. Apply the section to invented data

Items 1-5 use invented data for practice: a correlation matrix for university students. Items 6 and 7 are the book's exercises from Section 6.7.

	Sleep (hours)	Recall (words)	Stress	Caffeine (cups/day)
Sleep	—			
Recall	+.40	—		
Stress	−.45	−.35	—	
Caffeine	−.30	−.10	+.25	—

1. Which pair of variables has the strongest correlation, and which the weakest?
2. Describe the correlation between sleep and recall in words.
3. Controlling for caffeine, the partial correlation between sleep and recall is +.39. Interpret it as the book does.
4. Controlling for stress, it is +.29. Interpret it.
5. Why can you still not conclude that sleep improves memory?
6. Practice (Section 6.7): "Construct a correlation matrix for a hypothetical study including the variables of depression, anxiety, self-esteem, and happiness. Include the Pearson's *r* values that you would expect."
7. Discussion (Section 6.7): "Imagine a correlational study that looks at intelligence, the need for cognition, and high school students' performance in a critical thinking course. A multiple regression analysis shows that intelligence is not related to performance in the class but that the need for cognition is. Explain what this study has shown in terms of what is related to good performance in the critical thinking course."

3. Language exercise

Complete each interpretation.

1. If the partial correlation stayed almost the same (+.34), this would suggest that ...
2. If it dropped to almost zero (+.03), this would suggest that ...

3. If it fell from +.35 to +.20, this would suggest that …
4. Begin with "It is important to note that …" and state one limitation of factor analysis.
5. Correct the sentence: "If stress would explain the relationship, the partial correlation will drop to zero."

Speak with your tutor

1. In PSYC 110 Lecture 13, Professor Bloom criticises the family-meals study because the researchers "didn't factor out age" (the teenagers were 12 to 17). Explain to your tutor how partial correlation or multiple regression could control for age, and what problem would remain.
2. Lecture 13 also presents the Big Five ("OCEAN") and the idea of "G", a general factor found because people's scores on different intellectual tests are related. Ask your tutor how this connects with factor analysis, and compare it with the book's two factors of mathematical and verbal intelligence.
3. Explain the income, health and happiness example to your tutor, using *over and above* and *statistically controlled*.
4. Read the invented matrix aloud and interpret it with *would suggest that*. Ask for correction.

Writing task

Research project: adding statistical control to your correlational alternative (the second step of Lessons 17-18).

Extend your Lesson 17 text with a paragraph of 200-250 words. Include:

- two or three possible third variables (for example stress, caffeine, workload or part-time work) and how each will be measured;
- an empty correlation matrix (a table with your variable names, no numbers) that you will complete with the data;
- your analysis plan: a partial correlation between sleep and recall controlling for one variable, or a multiple regression with sleep and the third variables as predictors of recall;
- what two different results would suggest (conditional sentences);
- the limits that remain (directionality; unmeasured third variables).

Language: *if* + past, *would suggest that*; *statistically controlling for*; *over and above*; *It is important to note that*.

If you work on your own topic, add the same parts to your own correlational alternative.

Self-editing checklist:

- Each third variable has a clear reason and a measure.
- The matrix has every variable in both rows and columns.
- Predictor and outcome variables are named correctly.
- At least two possible results are interpreted.

- No *would* in an *if* clause.
- Causal language is avoided in the conclusions.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 13, Why Are People Different? (the Big Five, "G", and correlations in family studies): https://oyc.yale.edu/psychology/psyc-110/lecture-13

Answer key

1. Reading questions

1. They were healthier (more exercise, lower blood pressure), knew about heart attack risk factors, and correctly believed their own risk to be lower than their peers' ("Assessing Relationships Among Multiple Variables", paragraph 1).
2. The physical aggression measures tended to give highly skewed distributions (paragraph 2).
3. −.27; the weakest is between intelligence and social desirability (+.02) (correlation matrix table).
4. It organises many variables into a smaller number of clusters, strongly correlated within each cluster and weakly between them. Warmth, gregariousness, activity level and positive emotions correlate highly and represent extraversion ("Factor Analysis").
5. *Reflective and Complex*, .83. Pop: .59 on *Upbeat and Conventional* and .45 on *Energetic and Rhythmic* (music table).
6. Factors are not categories (they operate independently), and factor analysis only reveals the structure: researchers must interpret, label and explain the factors (paragraph after the table).
7. It does not solve the directionality problem, and unmeasured third variables may still drive the relationship ("Exploring Causal Relationships", last sentence).
8. How much the outcome changes for each one-unit change in a predictor. Multiple regression shows whether a predictor contributes over and above the other predictors ("Regression").

2. Apply the section to invented data

1. Strongest: sleep and stress (−.45). Weakest: recall and caffeine (−.10).
2. A positive relationship between the medium (.30) and large (.50) guidelines: students who sleep more tend to recall more words.
3. The relationship is largely independent of caffeine; caffeine is not a third variable driving it.
4. Stress accounts for some, but not all, of the relationship between sleep and recall.
5. Directionality remains (for example, students with better memory may finish studying earlier and sleep more), and unmeasured third variables, such as workload, may drive the relationship. An experiment is needed.
6. (example answer; other plausible values are correct)

	Depression	Anxiety	Self-esteem	Happiness
Depression	—			
Anxiety	+.60	—		
Self-esteem	−.50	−.40	—	
Happiness	−.60	−.40	+.50	—

Depression and anxiety should correlate positively with each other and negatively with self-esteem and happiness, which should correlate positively with each other.

7. (example answer) With both predictors in the analysis, need for cognition predicts performance over and above intelligence: students who like and value thinking tend to do better in the course, whatever their intelligence. Intelligence adds nothing once need for cognition is statistically controlled (on its own it might still correlate with performance, because the two predictors are themselves related). As the study is correlational, it shows what is related to good performance, not what causes it.

3. Language exercise

1. … the relationship is largely independent of SES (SES is not a third variable driving it).
2. … SES is indeed a third variable driving the relationship.
3. … SES accounts for some, but not all, of the relationship.
4. (example answer) It is important to note that factor analysis reveals only the underlying structure of the variables; researchers must interpret and label the factors themselves.
5. If stress explained the relationship, the partial correlation would drop to (almost) zero.

My notes